

Crime Records of India

Backend API Documentation

Enterprise Backend Documentation for Frontend Integration

1. Project Overview

Crime Records of India is a scalable backend platform built using Node.js, Express.js, MySQL, Redis, Docker, and JWT Authentication. The backend provides dynamic dataset APIs, analytics APIs, governance APIs, admin audit logging, Redis caching, and authentication flows.

2. Base URL

Environment	Base URL
Local Development	http://localhost:3000/api

3. Authentication

All protected APIs require JWT Authentication. Frontend must send access token in Authorization header.

Authorization: Bearer ACCESS_TOKEN

3.1 User Login API

Endpoint: POST /auth/login

Request Body

```
{
  "email": "test@test.com",
  "password": "User@123"
}
```

Sample Response

```
{
  "success": true,
  "accessToken": "JWT_ACCESS_TOKEN",
  "refreshToken": "JWT_REFRESH_TOKEN",
  "user": {
    "id": 1,
    "username": "testuser",
    "email": "test@test.com"
  }
}
```

3.2 Admin Login API

Endpoint: POST /auth/login-admin

```
{
  "username": "superadmin",
  "password": "Admin@123"
}
```

4. Dataset APIs

API	Method	Description
/tables	GET	Get all available dataset tables
/tables/:table/schema	GET	Get schema of dataset table
/data/:table	GET	Get paginated table data
/data/:table/search	GET	Search dataset records
/data/:table/filter-options/:column	GET	Get distinct filter options

4.1 Get Table Data

Endpoint: GET /data/:table?page=1&limit;=10

Example

GET /data/auto_theft?page=1&limit=10

Sample Response

```
{
  "success": true,
  "table": "auto_theft",
  "page": 1,
  "limit": 10,
  "data": [
    {
      "Area_name": "Delhi",
      "Year": 2001,
      "Auto_Theft_Stolen": 3185
    }
  ]
}
```

5. Analytics APIs

Endpoint: GET /analytics/overview

Provides aggregated analytics for dashboard and landing page.

```
{
  "success": true,
  "totalUsers": 20,
  "totalAdmins": 1,
  "totalTables": 17,
  "totalViews": 102,
  "totalSearches": 48
}
```

6. Admin Governance APIs

API	Method	Purpose
/admin/table/:table/update	PUT	Update dataset rows
/admin/table/:table/delete	DELETE	Delete dataset rows

Sample Update Request

```
{
  "conditions": {
    "Area_name": "Delhi",
    "Year": 2001
  },
  "newData": {
    "Auto_Theft_Stolen": 9999
  }
}
```

7. Redis Cache Layer

API	Redis Cached	TTL
/tables	Yes	30 Minutes
/tables/:table/schema	Yes	1 Hour
/analytics/overview	Yes	5 Minutes

8. HTTP Status Codes

Code	Meaning
200	Success
201	Created Successfully
400	Bad Request

401	Unauthorized
403	Forbidden
404	Resource Not Found
500	Internal Server Error

9. Frontend Integration Notes

- Store accessToken securely after login.
- Send Authorization header in protected APIs.
- Use pagination for large dataset tables.
- Schema APIs should be used for dynamic table rendering.
- Analytics APIs are Redis cached for performance.
- Admin APIs require admin JWT token.